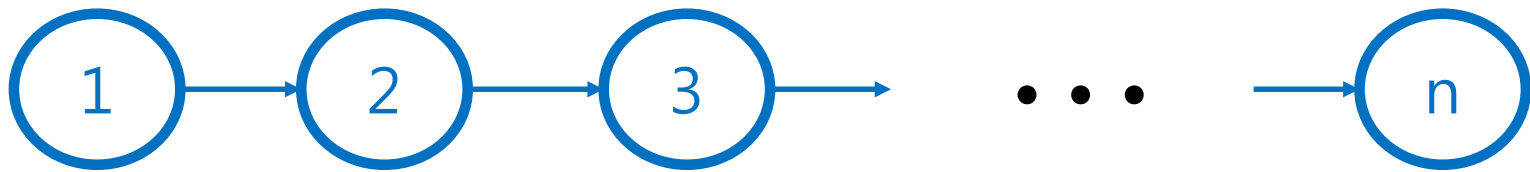


Project scheduling with multiple agents

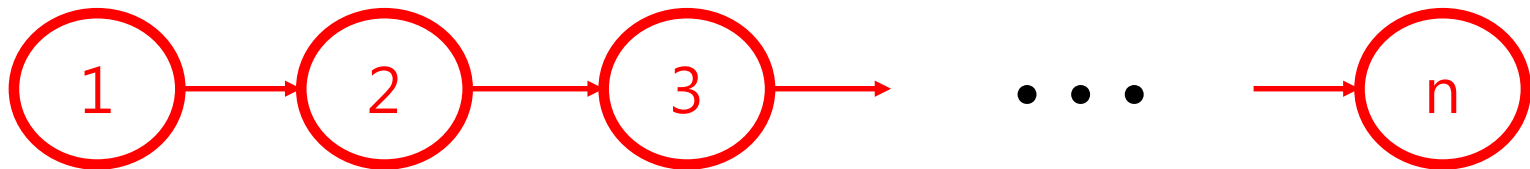
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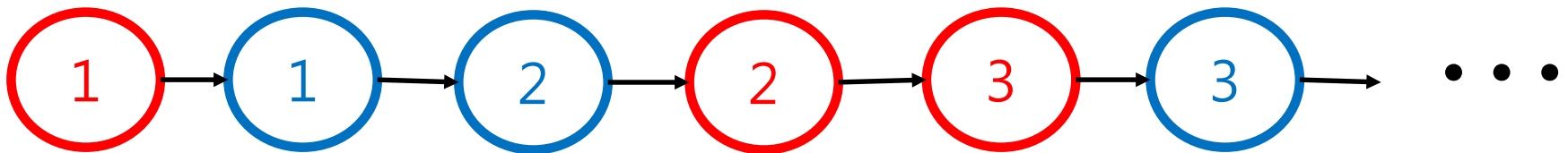
Motivation



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Previous research

1. Project scheduling on a chain graph

Product development process [1]

2. Agent scheduling [2]

Both agents have their own sets of jobs and share common resources for processing jobs with other agents. Furthermore, both agents are selfish and try to optimize their own objectives for their sets of jobs

Notation

Let $\sigma = (\pi, x)$ be the schedule such that

- Let $\pi = \left(\pi(1), \pi(2), \dots, \pi(\sum_{h=0}^1 n_h) \right)$ be the sequence, where $\pi(g)$ is the g th job. Note that precedence relation (2) between jobs belonging to the same project should hold in any sequence;

$$\text{Project } h : J_1^h \rightarrow J_2^h \rightarrow \dots \rightarrow J_{n_h}^h, \quad (2)$$

where $J_{j'}^h \rightarrow J_j^h$ represents that job J_j^h can start after job $J_{j'}^h$ is completed.

- Let $x = (x_1^1, x_2^1, \dots, x_{n_l}^l)$ be the compression vector, where x_j^i is the compressed amount of J_j^i . Note that the following inequalities should hold.

$$0 \leq x_j^h \leq u_j^h \quad \text{for } h = 0, 1 \quad \text{and } j = 1, 2, \dots, n_h.$$

Problem definition : Problem P1

$$\min z_1(\sigma) = C_{\max}^0(\sigma) + \sum_{j=1}^{n_0} c_j x_j$$

$$\text{s.t. } \sum_{j=n_0+1}^n c_j x_j \leq \mathcal{B},$$

$$C_{\max}^h(\sigma) \leq \mathcal{D}_h \quad \text{for } h = 1, 2, \dots, l,$$

$$0 \leq x_j \leq u_j \quad \text{for } j = 1, 2, \dots, n$$

Problem definition : Problem P2

$$\min z_2(\sigma) = \sum_{j \in T(\sigma)} w_j + \sum_{j=1}^{n_0} c_j x_j$$

$$\text{s.t. } \sum_{j=n_0+1}^n c_j x_j \leq \mathcal{B},$$

$$C_{\max}^h(\sigma) \leq \mathcal{D}_h \quad \text{for } h = 1, 2, \dots, l,$$

$$0 \leq x_j \leq u_j \quad \text{for } j = 1, 2, \dots, n,$$

where $T(\sigma)$ is the set of tardy milestones of project 0 in σ .

Problem P1

k-schedule

Let $\sigma_k = (\pi_k, x)$ be the schedule where project 0 is processed between projects k and $k + 1$, referred to as *k-schedule*. Note that

- In $k = 0$ ($k = l$), project 0 is processed first (last);
- In a k -schedule, the sequence of jobs is fixed.

To find an optimal k -schedule $\sigma_k^* = (\pi_k, x^*)$, we need only determine the compression amount of each job. We will seek σ_k^* using an approach based on linear programming(LP) whose idea is to reduce the problem to LP with a constant number of decision variables.

Optimality property 1

Lemma 1 *There exists an optimal k -schedule σ_k^* such that the jobs of each agent are compressed in non-decreasing order of unit compression cost.*

Observation 1 *Let $\rho_1 = (\rho_1(1), \rho_1(2), \dots, \rho_1(n_1))$ be the job sequence of project 1 in non-decreasing order of unit compression cost and let ω be an index such that*

$$\sum_{j=1}^{\omega-1} u_{\rho_1(j)} < \sum_{j \in V_1} p_j - \mathcal{D}_1 \leq \sum_{j=1}^{\omega} u_{\rho_1(j)}.$$

If a job $\rho_1(\omega)$ does not exist, then Problem P1 is infeasible. Otherwise, there exists an optimal k -schedule σ_k^ such that*

$$x_{\rho_1(j)}^* = u_{\rho_1(j)} \quad \text{for } j = 1, 2, \dots, \omega - 1 \quad \text{and} \quad x_{\rho_1(\omega)}^* \geq \sum_{j \in V_1} p_j - \mathcal{D}_1 - \sum_{j=1}^{\omega-1} u_{\rho_1(j)}.$$

Algorithm A

Procedure $A(k)$

Step 1 Set $t = 1$ and $\bar{V}_0 = \emptyset$.

Step 2 Set

$$\bar{p}_j = p_j, \quad \bar{u}_j = u_j, \quad \bar{x}_j = 0 \quad \text{for } j \in V_t, \quad \bar{V}_t = \bar{V}_{t-1} \cup V_t, \quad \text{and } \bar{n}_t = |\bar{V}_t|.$$

Step 3 Set $\rho_t = (\rho_t(1), \rho_t(2), \dots, \rho_t(\bar{n}_t))$ as the sequence of jobs in $\bar{V}_t$ in non-decreasing order of unit compression cost.

Step 4 If $\sum_{j \in \bar{V}_t} \bar{p}_j \leq \mathcal{D}_t$, then go to Step 8.

Step 5 Find a job $\rho_t(\omega_t)$ such that

$$\sum_{j=1}^{\omega_t-1} \bar{u}_{\rho_t(j)} < \sum_{j \in \bar{V}_t} \bar{p}_j - \mathcal{D}_t \leq \sum_{j=1}^{\omega_t} \bar{u}_{\rho_t(j)}. \quad (3)$$

If no job $\rho_t(\omega_t)$ exists that satisfies inequalities (3), then STOP (Problem P1 is infeasible).

Step 6 Let

$$\bar{x}_{\rho_t(j)} = \begin{cases} \bar{x}_{\rho_t(j)} + \bar{u}_{\rho_t(j)} & \text{for } j = 1, 2, \dots, \omega_t - 1 \\ \bar{x}_{\rho_t(j)} + \sum_{j \in \bar{V}_t} \bar{p}_j - \mathcal{D}_t - \sum_{j=1}^{\omega_t-1} \bar{u}_{\rho_t(j)} & \text{for } j = \omega_t. \end{cases}$$

Step 7 Let

$$\bar{p}_{\rho_t(j)} = \bar{p}_{\rho_t(j)} - \bar{x}_{\rho_t(j)} \quad \text{and} \quad \bar{u}_{\rho_t(j)} = \bar{u}_{\rho_t(j)} - \bar{x}_{\rho_t(j)}, \quad j = 1, 2, \dots, \omega_t.$$

Step 8 Set $t = t + 1$. If $t \leq k$, then go to Step 2.

Step 9 If $\sum_{j \in \bar{V}_k} \bar{c}_j x_j > B$, then STOP (Problem P1 is infeasible). Otherwise,

Optimality property 2

Observation 2 *In $\sigma_{k,f}^*$, the problem to determine the compression amount of the jobs in the projects in $\{f+1, f+2, \dots, l\}$ is independent of how to compress the jobs in the projects in $\{0, 1, \dots, f\}$.*

From Observation 2, we can develop a procedure similar to Procedure $A(k)$ that finds the compression amount of the jobs in the projects $\{f+1, f+2, \dots, l\}$ in $\sigma_{k,f}^*$. For simplicity, let this approach be referred to as *Procedure $B(f)$* and let $\mathcal{C}$ be the total compression cost of a schedule obtained by Procedure $B(f)$. Thus, we henceforth focus on how to determine the compression amount of the jobs in $\bigcup_{h=0}^f V_h$ under the state that the jobs of the projects in $\{f+1, f+2, \dots, l\}$ have been compressed by the amount obtained through Procedure $B(f)$. Note that this problem is expressed by LP as follows:

Reduced problem

$$\min \sum_{j=1}^{\hat{n}_k} (p_j - x_j) + \sum_{j=1}^{n_0} c_j x_j$$

$$\text{s.t. } \sum_{j=n_0+1}^{\hat{n}_f} c_j x_j \leq \mathcal{B} - \mathcal{C},$$

$$\sum_{j=1}^{\hat{n}_q} (p_j - x_j) \leq \mathcal{D}_q, \quad \text{for } q = k + 1, \dots, f - 1,$$

$$\sum_{j=1}^{\hat{n}_f} (p_j - x_j) = \mathcal{D}_f,$$

$$0 \leq x_j \leq u_j, \quad \text{for } j = 1, \dots, \hat{n}_f.$$

Optimality property 3

Lemma 3 *There exists an optimal schedule $\sigma_{k,f}^*$ for $LP(\alpha, \beta, \gamma)$ such that, for $j \in V_0 \setminus \{\alpha\}$*

$$x_j^* = \begin{cases} u_j, & \text{if } c_j < c_\alpha, \\ u_j, & \text{if } c_j = c_\alpha \text{ and } j \rightarrow \alpha, \\ 0, & \text{otherwise,} \end{cases}$$

for $j \in \bar{V}_k \setminus \{\beta\}$

$$x_j^* = \begin{cases} u_j, & \text{if } c_j < c_\beta, \\ u_j, & \text{if } c_j = c_\beta \text{ and } j \rightarrow \beta, \\ \bar{x}_j, & \text{otherwise,} \end{cases}$$

where $\bar{x}_j$ is the compression amount of job j obtained by Procedure $A(k)$ and for $j \in \bar{V}_f \setminus \{\gamma\}$

$$x_j^* = \begin{cases} u_j, & \text{if } c_j < c_\gamma \\ u_j, & \text{if } c_j = c_\gamma \text{ and } j \rightarrow \gamma \\ 0, & \text{otherwise.} \end{cases}$$

Strong polynomiality of Problem P1

Based on Lemma 3, the values of the decision variables in $LP(\alpha, \beta, \gamma)$ are determined, except for x_α, x_β , and x_γ . Then, $LP(\alpha, \beta, \gamma)$ is reduced to the LP with three decision variables, which implies strong polynomiality.

Theorem 1 *Problem P1 can be solved in strongly polynomial time.*

Problem P2

Complexity

In this section, we show that Problem P2 is NP-hard, even if the jobs belonging to project 0 cannot be compressed, that is, $u_j = 0$, $j = 1, 2, \dots, n_0$. Furthermore, we develop an efficient approach to the case in which we have no budget for compressing the jobs belonging to the projects in $\{1, 2, \dots, l\}$, that is, $\mathcal{B} = 0$.

For the proof of NP-hardness, we use the partition problem, which is known to be NP-complete [19]. The partition problem can be stated as follows: Given g positive integers $a_1, a_2, \dots, a_g$ whose sum is A , is there a subset $I \subset \{1, 2, \dots, g\}$ such that $\sum_{j \in I} a_j = A/2$?

Theorem 2 *Problem P2 is NP-hard, even if $u_j = 0$, $j = 1, 2, \dots, n_0$.*

Special case

Henceforth, we show that Problem P2 with $\mathcal{B} = 0$ can be solved in strongly polynomial time. Since the jobs belonging to the projects in $\{1, 2, \dots, l\}$ cannot be compressed, it is observed from Proposition 1 that project h can be regarded as one job, denoted job $n_0 + h$, $h = 1, 2, \dots, l$. For consistency, furthermore, let $\mathcal{D}_h$ be referred to as *due-date* of job $n_0 + h$, denoted d_{n_0+h} . By Proposition 2, furthermore, we consider only schedules satisfying

$$n_0 + 1 \rightarrow n_0 + 2 \rightarrow \cdots \rightarrow n_0 + l.$$

Optimality properties

Lemma 4 *The optimal schedule satisfies one of the following condition:*

- *All the jobs are uncompressed;*
- *There exists at least one JIT job.*

By Lemma 4, the compression amount of each job can be determined if an optimal schedule has no JIT job. Henceforth, we assume that there exists a JIT job in an optimal schedule. Let the schedule with $(y, \mathcal{I}, z)$ -property be such that y and z are consecutive JIT jobs and jobs in $\mathcal{I}$ are processed between y and z . Henceforth, under the assumption that an optimal schedule with $(y, \mathcal{I}, z)$ -property exists, we present a two-phase approach to find the subschedule for the jobs in $\mathcal{I} \cup \{z\}$.

Optimality properties

Lemma 5 *There exists an optimal schedule $\sigma^* = (\pi^*, x^*)$ with $(y, \mathcal{I}, z)$ -property such that the jobs in $\mathcal{I}^0$ are compressed in order of sequence τ .*

Lemma 6 *An optimal sequence of the jobs in $\mathcal{I}$ can be found in strongly polynomial time.*

Theorem 3 *Problem P2 with $\mathcal{B} = 0$ can be solved in strongly polynomial time.*

References

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2. A. Agnetis, P.B. Mirchandani, D. Pacciarelli, A. Pacifici, Scheduling problems with two competing agents, Operations Research 52 (2004) 229--242.